

COVID-19 Global Data Analysis Report

DATA SCIENCE PORTFOLIO PROJECT | MAY 2026

1. Objective

This report provides a data-driven overview of the COVID-19 pandemic's impact globally. The analysis focuses on three primary metrics: cumulative infection rates, mortality statistics across continents, and the percentage of population affected in specific jurisdictions. The goal is to highlight regional disparities and overall global health trends.

2. Global Summary Metrics

Based on the consolidated data from the initial project phase, the global impact of the virus is summarized below:

150,574,977	3,180,206	2.11%
TOTAL CONFIRMED CASES	TOTAL DEATHS	GLOBAL FATALITY RATE

3. Regional Mortality Analysis

The burden of mortality has been unevenly distributed across the globe. Continental data indicates that Europe and North America faced the highest absolute death counts recorded in this dataset.

Continent / Location	Total Death Count
Europe	1,016,750
North America	847,942
South America	672,415
Asia	520,269

Continent / Location	Total Death Count
Africa	121,784
Oceania	1,046

4. Population Infection Density

Analyzing the infection rate as a percentage of the total population reveals that smaller nations and territories often experienced higher densities of transmission. For instance, countries like Andorra and Montenegro saw over 15% of their total populations infected at various peaks of the pandemic.

In larger nations, the United States maintained one of the highest infection-to-population ratios at approximately 9.77%.

5. Technical Implementation

The analysis was conducted using a robust data pipeline involving:

- **SQL:** For data cleaning, joining "CovidDeaths" and "CovidVaccinations" tables, and performing aggregate calculations.
- **Tableau:** Utilized for the interactive dashboard (Covid Dashboard.twbx) which provides temporal visualizations of case surges.
- **Data Sources:** High-fidelity global datasets processed into structured tables (TBL_1 through TBL_4).

6. Conclusion

The analysis confirms that while the global fatality rate was relatively consistent at ~2.1%, the logistical and health burden varied immensely by geography. These insights are critical for future pandemic preparedness and for understanding the efficacy of public health interventions like vaccination drives, which are integrated into the interactive portion of this project.

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Project Repository: COVID-19-Portfolio-Analysis